Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 20.2

$$
\begin{aligned}
&\int \sinh 5x dx \\
&= \int \frac{e^{5x} - e^{-5x}}{2} dx \\
&= \frac{1}{2} \int \left(e^{5x} - e^{-5x}\right) dx \\
&= \frac{1}{2} \left(\int e^{5x} dx - \int e^{-5x} dx \right) \\
&= \frac{1}{2} \left(\frac{1}{5} e^{5x} - \left(-\frac{1}{5} e^{-5x} \right) \right) + C \\
&= \frac{1}{2} \left(\frac{1}{5} e^{5x} + \frac{1}{5} e^{-5x} \right) + C \\
&= \frac{1}{10} \left(e^{5x} + e^{-5x} \right) + C \\
&= \frac{1}{5} \cosh 5x + C
\end{aligned}
$$

References